



**Stan's
Technologies**

4D Cluster Density Analysis

Information Density in K-map Minimization

Analysis Date: 2026-01-17

Random Seed: 42

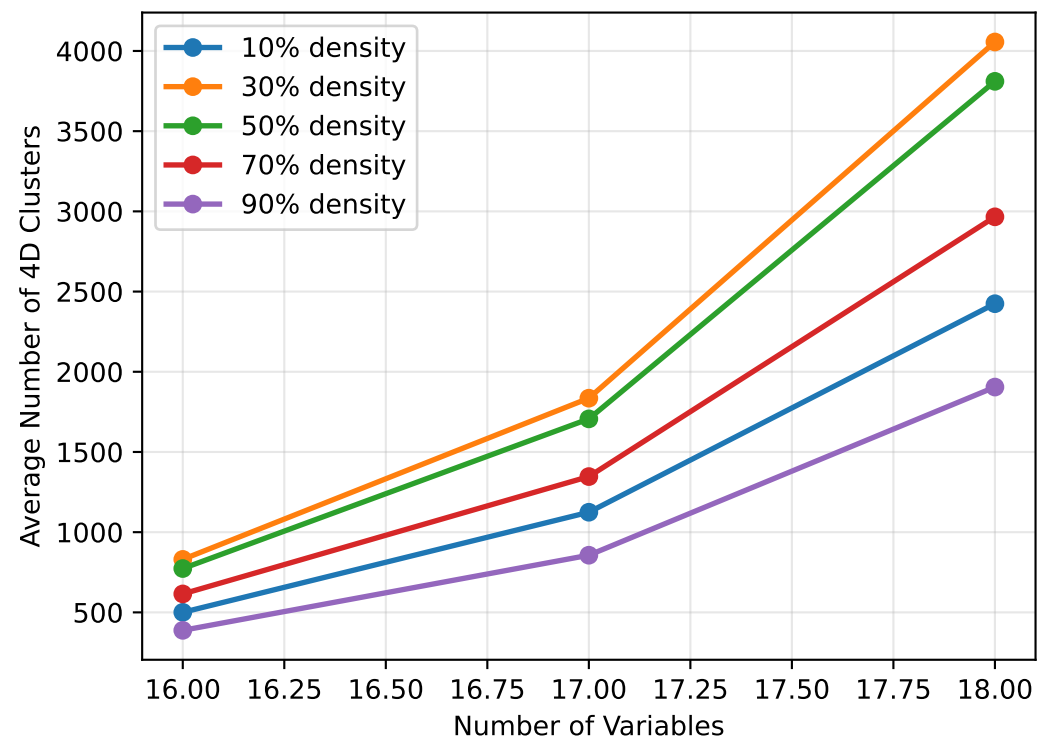
Total Test Cases: 45

Variable Range: 16-18

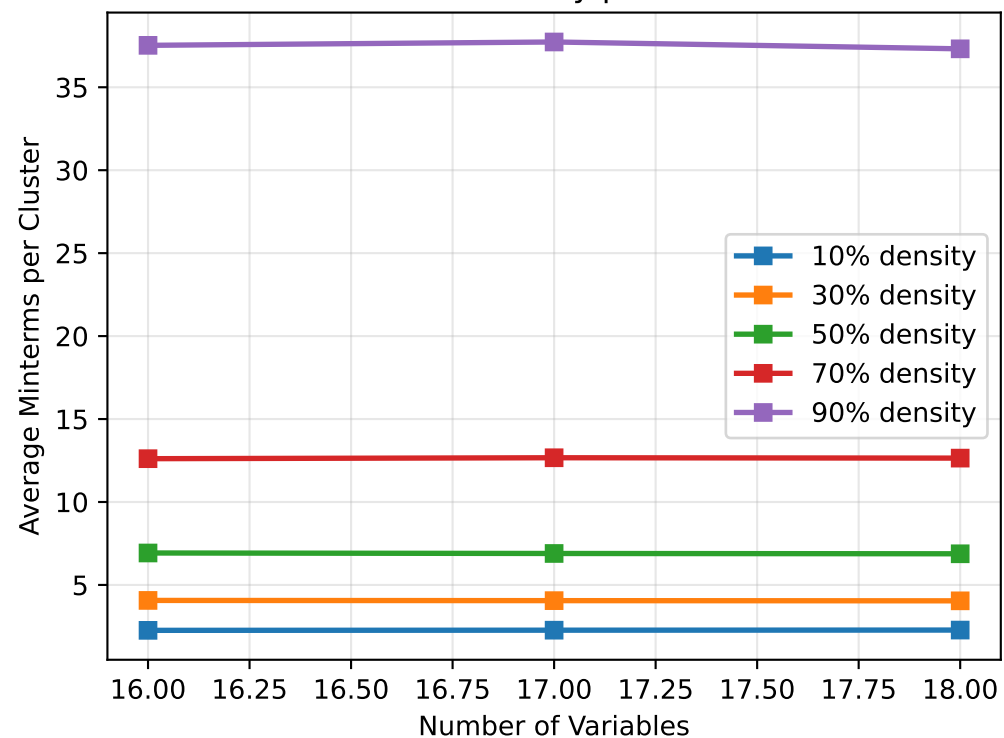
Geometric Clustering Behavior Analysis

© Stan's Technologies 2026

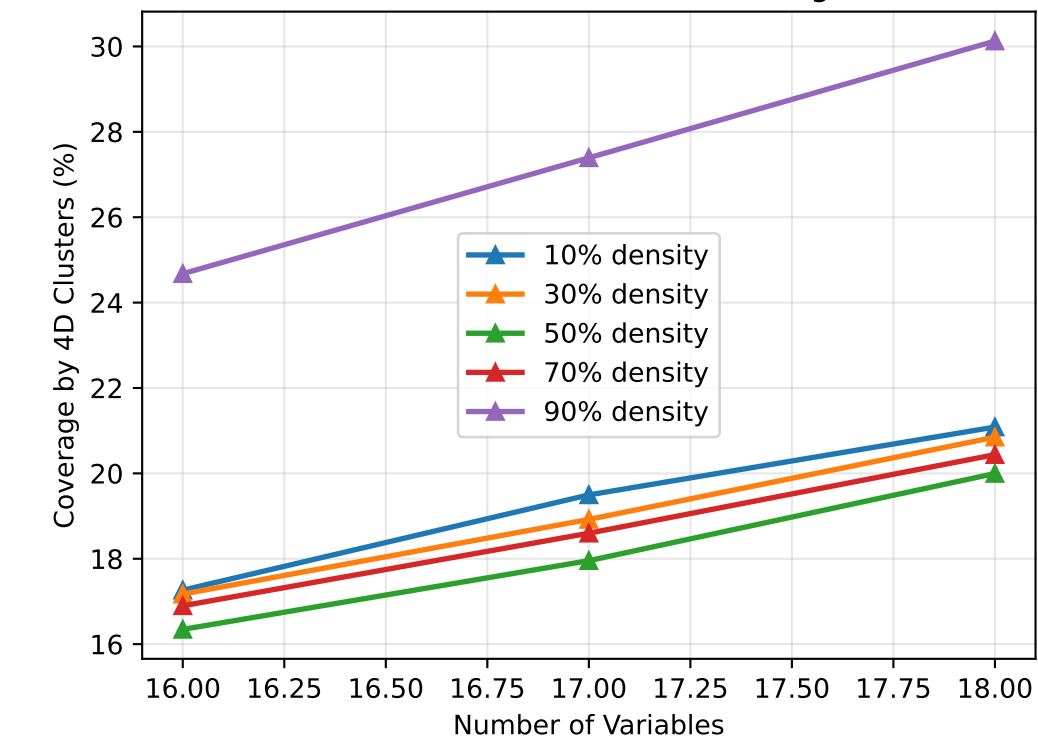
4D Cluster Formation vs Problem Size



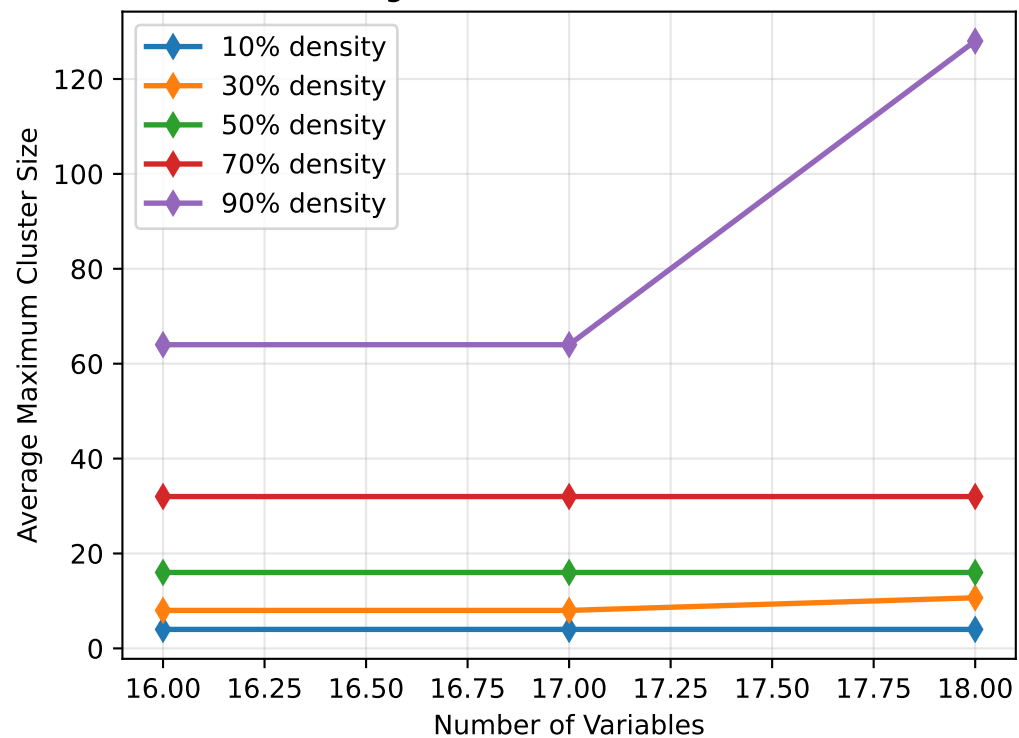
Information Density per 4D Cluster



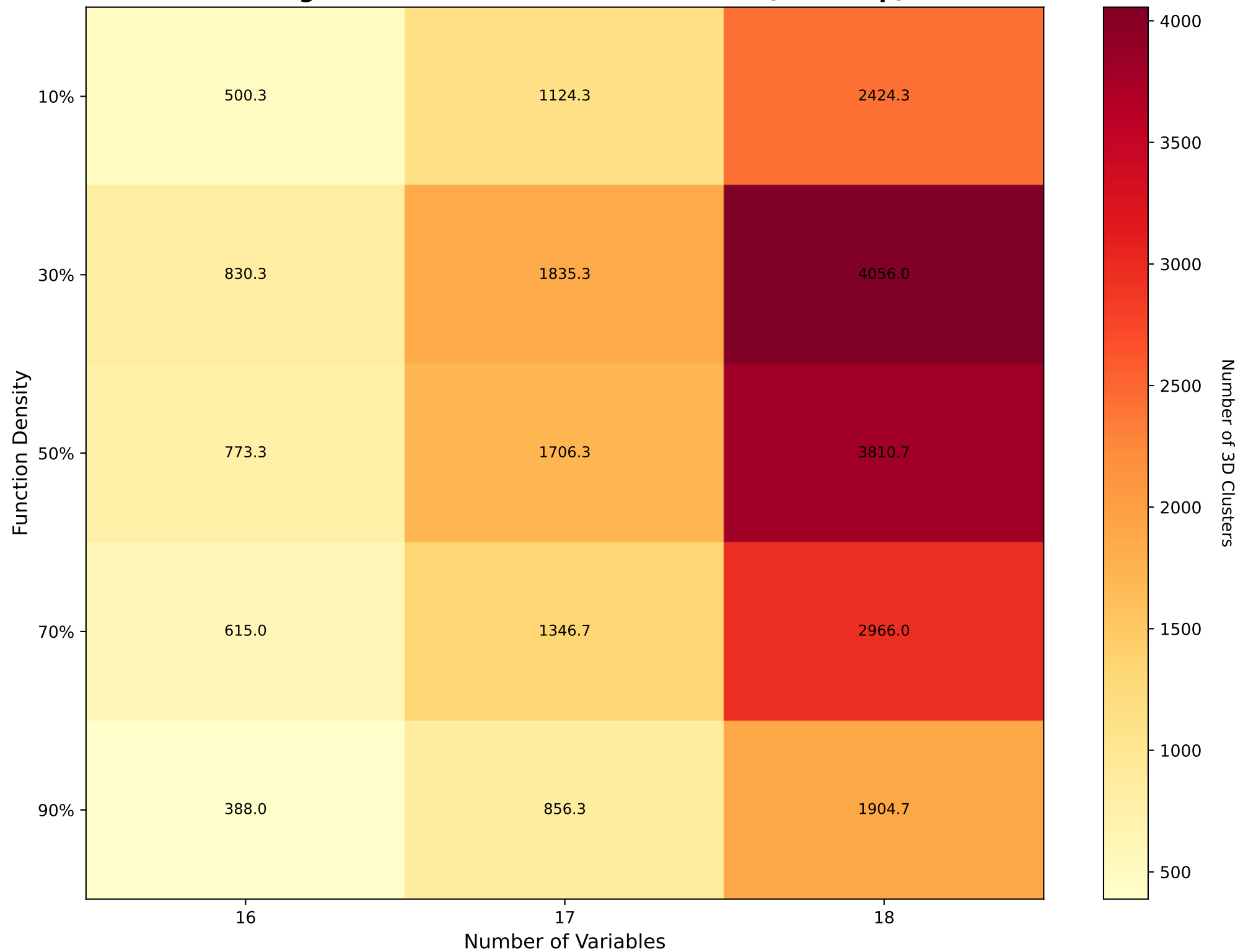
Effectiveness of 4D Clustering



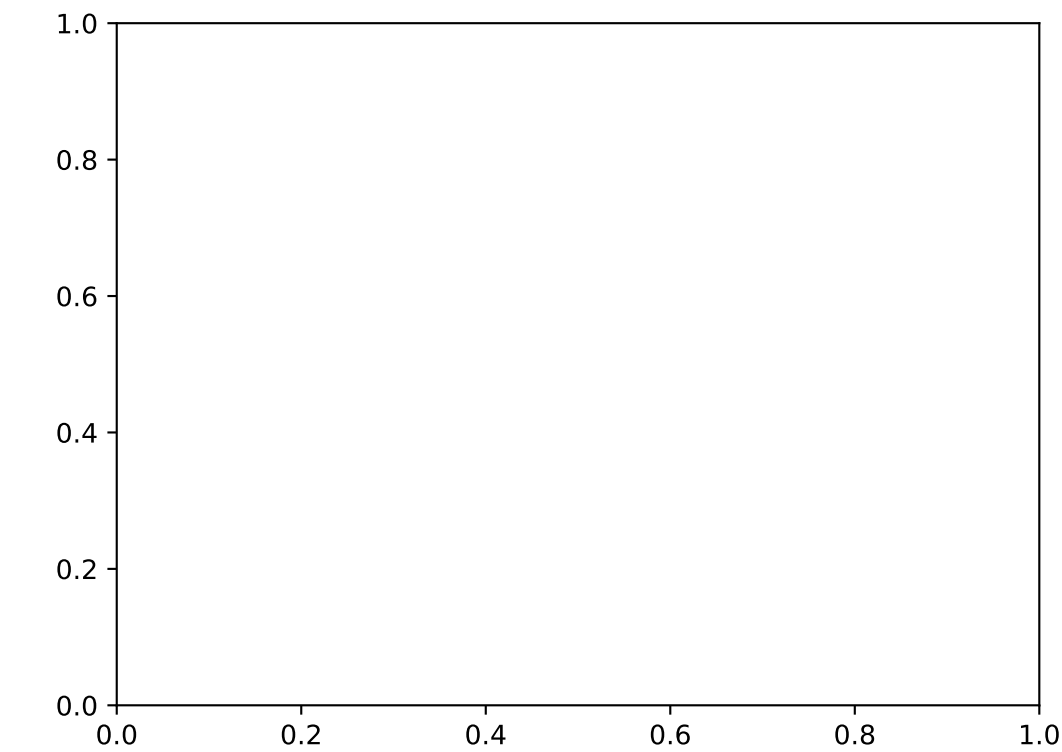
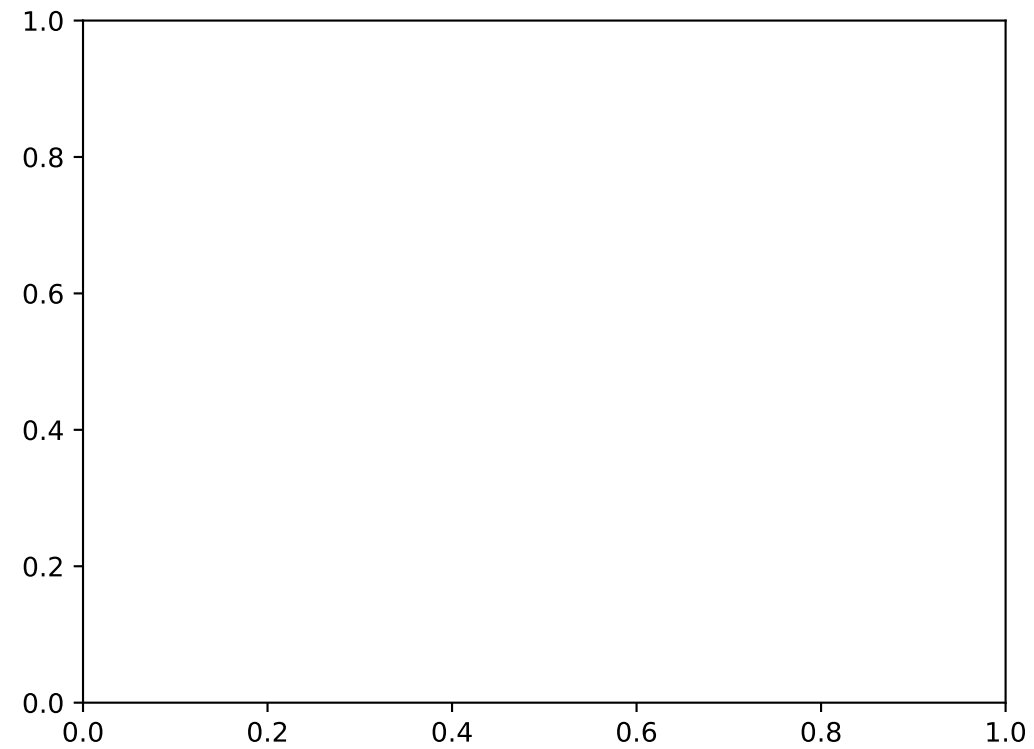
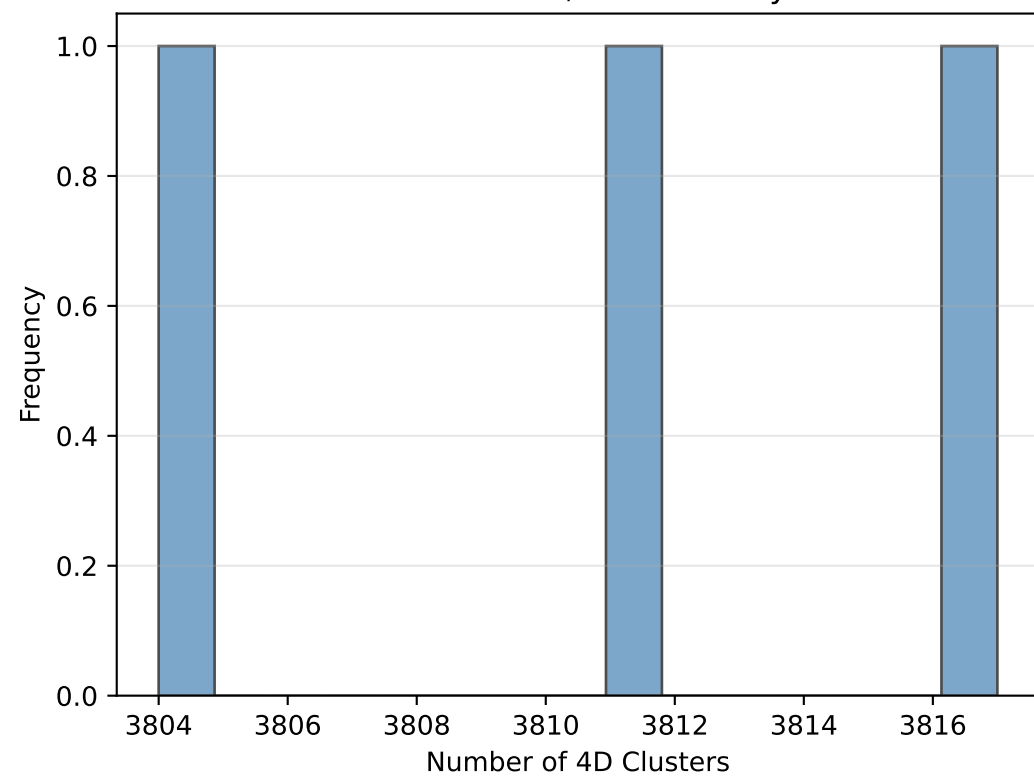
Largest 4D Cluster Formation



Average Number of 3D Clusters Formed\n(Heatmap)



18 Variables, 50% Density



Analysis Summary

Key Findings

- Total tests conducted: 45
- Average 4D clusters per test: 1675.84
- Average coverage by 4D clusters: 20.5%
- Variable range: 16 to 18
- Density levels tested: 5

Observations

- *4D cluster formation increases with variable count*
- *Higher density functions tend to form more 4D clusters*
- *Average cluster size varies with function structure*
- *Coverage by 4D clusters is significant in most cases*